

Every month, we're spending at least 60 euros on hobbies. Some of us pay another 20 to 30 euros for a Basic-Fit membership. But what's the point of all this spending?

2. The money we spend on gyms and hobbies is really about chasing that feeling. Staying fit gives us more than just muscles—it gives us a sense of independence. You can move on your own, live on your own terms, and not rely on others for help.

3. But there's something simple we're all forgetting. When you meet someone, you can instantly sense if they're young or old. Healthy or not. That first impression tells you a lot—and it's based on something very basic. And because it's so basic, we've stopped paying attention to it.

4. The one thing no one talks about enough: skin.

5. We all assume our skin will stay the same—tight, smooth, fresh—forever. But that's just not how it works. Look at your grandparents. Their skin tells a story. **Wrinkles, sagging, spots.** And if you ask them, many will say they wish they had taken better care of it.

6. Skin is what people notice first. It's the first thing that gives away your **age**, your **health**, your energy. It's also the **last thing you wear**. Not your shirt. Not your shoes. Your skin is your body's one true outfit—your first and last layer.

And here's what's wild: when people go in for cosmetic surgery, it's almost always about the skin. Not because it's damaged—but because it *shows* everything. Time, stress, neglect.

So while we spend money on gyms, hobbies, and fitness to stay young and capable, we're often forgetting the one part of our body that never lies: our skin.

7. So, how do we protect our skin?

The first step is simple: know your skin.

Once you understand your skin type, it becomes much easier to take care of it—whether that means keeping it smooth and fresh, or starting a treatment routine. The problem is, most of us walk into a cosmetic store, see hundreds of products, and have no idea which one will actually work for our skin.

That's exactly why I created this app.

With just a photo, the app scans your skin and tells you your skin type. Based on that, it recommends the right kind of products specifically for *you*. No guessing. No wasting money on things that don't work.

The app also has a subscription feature. If you choose it, you'll receive a personalized skincare package every month, designed to match your skin's needs. You might be wondering—why subscribe when you could just buy products yourself?

Here's the difference:

- You won't forget your routine.
- You won't waste time or money on the wrong products.
- You'll follow a process. Good skin doesn't come from using the same product over and over—it comes from *evolution*. Each month, the products in your package adjust based on updates you share in the app. That way, your skin gets what it actually needs as it changes.

This is skincare with direction—not guesswork.

8. So, in my project, I worked with four skin types: normal, dry, oily, and diseased. Since there are four classes, the baseline accuracy is 25%—meaning if the model just guessed randomly, it would be right one out of four times.

I collected around 500 images for each class to keep it balanced. I also made sure all the images were in the same format — I converted them to JPG, removed any that were blurry or not sharp, and filtered out any that weren't in RGB.

I did this to make sure the dataset was clean and consistent so the model could learn properly without being biased or confused by low-quality images.

9. As you all know, we did four iterations to train our dataset and test which model works best and gives us the best output. In iteration 3, I chose to use a pretrained model — specifically, VGG16. The reason I picked VGG16 is because, based on my research, it performs well on complex and large datasets, and it's known for extracting image features effectively. Since it's a pretrained model, I didn't need to train it from scratch, which also saves time and computational resources.

VGG16 requires input images to be 224x224 in size, because it was originally trained on images of that dimension. That's why I changed my image size from 256 to 224. Also, since I'm working with more than two classes, I used "categorical" as the class mode instead of "binary." I set the batch size to 16 so the model could learn more effectively from the dataset and also to help improve the final output.

I didn't apply data augmentation in this iteration. First, I wasn't allowed to, and second, when I tried it earlier, the results got worse. I realized that data augmentation might confuse the model — for example, it could transform an oily skin image in a way that makes it look like normal or diseased skin, which would affect the accuracy. That's why I avoided using it.

For the CNN architecture, as I mentioned, I used the pretrained VGG16 model. The input shape of the images was (224, 224, 3), which confirms all images were in RGB format. After the VGG16 base, I added a Flatten layer to convert the 3D output to 1D, and then used three dense layers to help the model learn better features. At the end, I used a **softmax** activation function in the output layer to classify the images into one of the four skin types.

10. During training, I used a learning rate of 0.00003, set accuracy as a metric, and used categorical crossentropy as the loss function since I had four classes. I also implemented **early stopping to stop training if the model started performing well within the first few epochs.**

11. As for the results, the VGG16 model achieved about 69% accuracy after 29 epochs and reached around 64% accuracy on the test data. Both the validation loss and test loss were close to 0.5, which I consider a decent result.

12. We all know that CNNs are powerful when it comes to image classification, but the problem is that they're usually considered a black box. We get predictions from the model, but we don't really know *why* it made that decision or *what part* of the image it was focusing on. For example, when the model says an image is "oily skin," we don't know what exactly it saw to make that choice.

That's where XAI, or explainable AI, comes in. In my project, I used a technique called Grad-CAM, which helps us "open the black box" and understand what's happening inside. Grad-CAM gives us a **heatmap** that shows which parts of the image the model paid the most **attention** to when making its prediction.

This is really helpful, especially in cases where the model doesn't make the right prediction. Using XAI, we can actually see why the model might have made a mistake — maybe it focused on the background instead of the skin, or it picked up the wrong features.

Another important thing about using explainability is that it builds **trust**. When people can see why the model predicted something — and it makes sense — they're more likely to trust the system. That's one of the reasons I used Grad-CAM. **It helps us visually understand what the model is seeing and gives us more confidence in its decisions.**

13. The feedback I got from my peers about my wireframe was quite positive overall. There were just a few small suggestions for improvement. One of the main points was adjusting the **font size** to make it more readable. Another suggestion was to move the **menu** from the top left to the bottom of the screen, since they were seeing it more like a website layout rather than a mobile app.

Other than that, they really liked the **layout and color scheme** of the application. Everyone agreed that the design was **simple, easy to use, and easy to understand**, even without showing them the persona or the task list.